

HSBI — Bielefeld University of Applied Sciences and Arts

Quantitative Research Methods

Summer Semester 2026 · Examiner: Prof. Dr. Christoph Weisser

Final Mock Examination — Set B (Lectures 1–12)

Comprehensive coverage of ISLP Chapters 2–8, 10 and 13,
with emphasis on Chapters 7 (splines/GAMs), 8 (trees), 10 (deep learning) and 13 (multiple testing)

Duration: 120 minutes

Total credit: 120 points

Allowed aids: non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: _____

Matriculation no.: _____

Problem	P1	P2	P3	P4	P5	P6	P7	Total
Maximum points	16	8	20	16	20	20	20	120
Achieved points								

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- All numerical constants you need (powers of e , logarithms, $(0.95)^{10}$, t -quantiles) are provided in the problems.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Moving beyond linearity — splines

(16 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

Throughout this problem, consider regression splines built on a single predictor X using $K = 5$ interior knots.

- (a) [4 P] A *cubic spline* is a piecewise cubic polynomial that is continuous and has continuous first and second derivatives at every knot. How many degrees of freedom (free parameters, counting the intercept) does a cubic spline with K interior knots use? Justify your answer by counting the parameters of the polynomial pieces and subtracting the constraints at the knots, and evaluate for $K = 5$.
- (b) [4 P] How many degrees of freedom does a *natural* cubic spline with the same $K = 5$ knots use? State the additional constraint a natural spline imposes, in which region of the predictor space it acts, and why practitioners find this constraint useful.
- (c) [4 P] A *smoothing spline* is the function g that minimises

$$\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt.$$

Explain the meaning of the two terms, and describe the shape of the solution $\hat{g}$ in the two limiting cases $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$, giving the effective degrees of freedom that $\hat{g}$ approaches in each case.

- (d) **[4 P]** Write down the *truncated-power basis* representation of a cubic spline with a single knot at ξ , define the truncated-power basis function $(x - \xi)_+^3$ explicitly, and explain in one sentence why adding this term keeps the fit continuous with continuous first and second derivatives at ξ .

Problem 2: Generalised additive models

(8 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

- (a) **[3 P]** Write down the general form of a *generalised additive model* (GAM) for a quantitative response Y and predictors $X_1, \dots, X_p$, and explain in one sentence how it generalises multiple linear regression.
- (b) **[3 P]** State one important *interpretability advantage* of GAMs (concerning the fitted functions f_j) and their central structural *limitation*, including how that limitation can be repaired.
- (c) **[2 P]** Describe in one or two sentences how the *backfitting* algorithm fits a GAM.

Problem 3: Regression and classification trees by hand

(20 points)

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

A logistics firm records, for $n = 6$ regional depots, the number of delivery vans x and the number of parcels delivered per day y (in thousands):

Depot i	1	2	3	4	5	6
Vans x_i	10	12	14	16	18	20
Parcels y_i	6	4	5	11	13	12

A regression tree with a *single* split at cutpoint s partitions the depots into $R_1 = \{i : x_i < s\}$ and $R_2 = \{i : x_i \geq s\}$ and predicts the region mean $\hat{y}_{R_m}$ in each region.

- (a) **[8 P]** For each of the two candidate cutpoints $s = 15$ and $s = 17$, determine the two regions, their means, and the total residual sum of squares $\text{RSS}(s) = \sum_{m=1}^2 \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2$. Which cutpoint does recursive binary splitting prefer, and why?
- (b) **[2 P]** Suppose recursive binary splitting selects the split at $s = 15$, giving regions $R_1 = \{x < 15\}$ with mean $\hat{y}_{R_1} = 5$ and $R_2 = \{x \geq 15\}$ with mean $\hat{y}_{R_2} = 12$. Predict the number of parcels for a new depot with $x = 13$ vans.
- (c) **[6 P]** A *classification* tree for a different task has a terminal node containing 12 training observations from three classes: 6 “Premium”, 3 “Standard” and 3 “Basic”. Compute the class proportions, the Gini index $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$ and the entropy $D = -\sum_k \hat{p}_k \ln \hat{p}_k$ of this node (use $\ln 0.5 = -0.6931$ and $\ln 0.25 = -1.3863$). Which class does the node predict?

- (d) [4 P] In *cost-complexity pruning* one minimises, over subtrees $T \subseteq T_0$ of the large tree T_0 ,

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|.$$

Explain the role of α : what happens for $\alpha = 0$, what happens as α grows, and how is α chosen in practice?

Problem 4: Ensembles — bagging, random forests, boosting (16 points)

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

- (a) [6 P] Let $\hat{f}_1(x), \dots, \hat{f}_B(x)$ be B identically distributed (but not independent) tree predictions at a fixed point x , each with variance σ^2 and pairwise correlation ρ . The variance of the bagged average is

$$\text{Var}\left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x)\right) = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2.$$

- (i) Explain briefly why this formula shows that bagging reduces variance and which term limits the reduction. (ii) Evaluate the variance for $B = 500$, $\rho = 0.7$ and $\sigma^2 = 4$, and compare it with the variance of a single tree. (iii) What value does the variance approach as $B \rightarrow \infty$?
- (b) [4 P] How do *random forests* modify bagging at each split, and what is the recommended subset size m for *classification* as a function of p (evaluate for $p = 36$)? Using the variance formula $\text{Var} = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2$ from part (a), explain why this modification improves on bagging.
- (c) [4 P] A data scientist fits a boosting model for regression trees in `scikit-learn`:

```
GradientBoostingRegressor(n_estimators=500, learning_rate=0.01, max_depth=1)
```

These three arguments are exactly the tuning parameters of boosting. State in one sentence each what `n_estimators`, `learning_rate` and `max_depth` control, and explain the “slow learning” idea behind the small `learning_rate`.

- (d) [2 P] What does the *out-of-bag* (OOB) error of a bagged model estimate, and why is it essentially free to obtain?

Problem 5: Neural networks by hand (20 points)

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Consider a small feed-forward network for a quantitative response with input $x = (x_1, x_2) = (2, 1)$, one hidden layer of two ReLU units ($g(z) = \max\{0, z\}$) and a linear output unit:

$$A_1 = g(-2 + 3x_1 - 1x_2), \quad A_2 = g(1 - 2x_1 + 1x_2), \quad f(x) = -1 + 2A_1 + 3A_2.$$

- (a) [8 P] Carry out the forward pass step by step: compute both pre-activations, both activations A_1, A_2 and the output $f(x)$. Comment briefly on what the ReLU did to the second unit and why non-linear activations are essential.

- (b) **[6 P]** A classification network with $K = 3$ classes returns the logits (output-layer pre-activations) $z = (z_1, z_2, z_3) = (2.0, 1.0, -1.0)$ for one observation whose true class is class 1. Compute all three softmax probabilities and the cross-entropy loss $-\ln \hat{p}_{\text{true}}$. (Use $e^2 = 7.389$, $e^1 = 2.718$, $e^{-1} = 0.368$ and $\ln 10.475 = 2.349$.)
- (c) **[4 P]** Count the parameters of a fully connected network with $p = 6$ inputs, one hidden layer of $k = 4$ units and an output layer of $K = 3$ units, where every hidden and output unit carries a bias. Show the count separately for the two weight layers.
- (d) **[2 P]** A convolutional layer receives a 32×32 image and applies filters of size $F = 5$ with padding $P = 0$ and stride $S = 1$. Compute the spatial size of the output feature map from $(W - F + 2P)/S + 1$.

Problem 6: Multiple testing

(20 points)

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

- (a) **[4 P]** A lab runs $m = 10$ independent hypothesis tests, each at level $\alpha = 0.05$, and suppose all 10 null hypotheses are true. Define the family-wise error rate (FWER) and compute it here (use $(0.95)^{10} = 0.5987$). Interpret the result in one sentence.
- (b) **[7 P]** A team screens $m = 6$ candidate biomarkers; the ordered p -values are

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.009	0.012	0.015	0.040	0.300

Control the FWER at $\alpha = 0.05$ (i) with *Bonferroni* and (ii) with the *Holm step-down* procedure. Give the relevant thresholds, state exactly which hypotheses each method rejects, and explain why Holm can never reject fewer hypotheses than Bonferroni.

- (c) **[6 P]** Apply the *Benjamini–Hochberg* (BH) procedure at FDR level $q = 0.05$ to the same six ordered p -values ($m = 6$; $p_{(1)}, \dots, p_{(6)} = 0.002, 0.009, 0.012, 0.015, 0.040, 0.300$): give the thresholds jq/m , determine which hypotheses are rejected, and state precisely what “FDR control at $q = 0.05$ ” guarantees for the resulting set of rejections.
- (d) **[3 P]** A pharmaceutical company runs a *confirmatory* Phase-III trial with $m = 4$ pre-specified primary endpoints; a single significant endpoint could support regulatory approval of the drug. Should the company control the FWER or the FDR? Justify your recommendation.

Problem 7: Cumulative essentials (Chapters 2–6)

(20 points)

Lecture slides: Chapters 2–6 (Lectures 1–8).

- (a) **[4 P]** A medical screening classifier is evaluated on 200 patients (label 1=diseased, 0=healthy). In `scikit-learn`, `confusion_matrix(y_true, y_pred)` returns the array below (rows = true class, columns = predicted class; label order $[0, 1]$):

	predicted 0	predicted 1
true 0 (healthy)	120	30
true 1 (diseased)	10	40

Read off the counts and compute the overall error rate, the sensitivity, the specificity and the precision (positive predictive value) for the positive class 1. Which single number would trace out the ROC curve if it were varied, and in which direction would sensitivity and specificity move?

- (b) **[4 P]** Consider K -nearest-neighbours. (i) Using the training set (x, y) : $(1, 3)$, $(4, 6)$, $(6, 8)$, $(7, 7)$, $(10, 12)$, predict the response at $x_0 = 5$ with $k = 3$ (KNN regression, unweighted average). (ii) Explain how k governs the bias–variance trade-off, which extreme ($k = 1$ or k large) is the most flexible, and why the *training* error of 1-NN is always 0.
- (c) **[4 P]** Explain in one or two sentences what k -fold cross-validation does, then give one reason to prefer $k = 5$ or 10 over *LOOCV* and one reason to prefer it over a *single* validation split.
- (d) **[2 P]** You have $p = 2,000$ predictors but expect only a handful to be truly related to the response. Ridge or lasso — which penalty is more appropriate, and where would principal-components regression (PCR) sit relative to your choice?
- (e) **[6 P]** Match each scenario to exactly one method from $\{\textit{linear regression}, \textit{lasso}, \textit{logistic regression}, \textit{random forest}\}$ (each method used exactly once) and justify each choice in one sentence:
 - (i) Estimate how much one additional year of schooling adds to annual earnings, with a confidence interval, from $n = 5,000$ survey responses.
 - (ii) Predict a continuous blood-pressure outcome from $p = 8,000$ genetic markers for $n = 150$ patients, and report a short list of the relevant markers.
 - (iii) Classify each incoming e-mail as spam or not, and explain to auditors how each feature changes the odds of an e-mail being spam.
 - (iv) Predict machine failure from hundreds of mixed sensor variables with strong non-linear effects and interactions; only predictive accuracy matters.

End of the examination — good luck!